# Data Safety Training

Bootcamp edition · Safeguarding data in Vector's AI environment

VECTOR INSTITUTE | INSTITUT VECTEUR

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INTRODUCTION

# Why this course exists

Course objective

## Give every participant the knowledge and skills to safeguard data, in line with best practice and with Canadian & Ontario privacy law.

Fostering a culture of data stewardship and privacy awareness across Vector's AI discovery settings.



WHO MUST COMPLETE IT

- Vector staff
- Faculty & Faculty Affiliates
- Distinguished postdoctoral fellows
- Graduate students & postdocs
- Research interns
- Project collaborators

*Required before access to Vector's Data & Analytics Environments is granted.*

INTRODUCTION

# The Data & Analytics Environment (DAE)

Vector's entire computing capability supporting the community: Vector-controlled facilities, cloud services, and the federal PAICE compute infrastructure.

Using the DAE means complying with applicable laws, frameworks and the specific obligations set in data agreements, terms & conditions, and licenses.



### Limits on user activity

e.g. no commercial use; no attempts at re-identification



### Technical & administrative safeguards

e.g. user authentication; secure, access-controlled facilities

INTRODUCTION

# Why compliance matters for ML / AI



### Data is the fuel

Large datasets are essential to ML / AI research.



### The law protects people

Canada and other countries have privacy laws shielding individuals and their data.



### Know your obligations

Participants must understand the legal requirements that apply to them.

**Sensitive data may need more.** Access to sensitive or de-identified health data may require additional training (TCPS 2, CITI). Vector's Data Office will advise as needed.

# Every dataset must be de-identified

## NO EXCEPTIONS

Vector does not hold the legal clearances required to keep identifiable data. All data entering the DAE must be de-identified first.

### What de-identification means

Removing any information that identifies an individual, or that could reasonably be used, alone or combined with other data, to identify them.



**Your responsibility**

Participants bringing data in must ensure it is de-identified and meets Vector's De-Identification Guidelines before transfer to the DAE.

# Direct vs. indirect identifiers

DIRECT IDENTIFIERS

## Identify a person on their own

- Full name
- Health card / medical record no.
- Social insurance number
- Email & phone

INDIRECT (QUASI) IDENTIFIERS

## Re-identify when combined

- Date of birth
- Gender
- Postal code / location
- Marital status, job, ethnicity

# De-identification has limits

## Risk never reaches zero

As public datasets multiply, re-identification risk grows. Linking datasets, even de-identified ones, often raises that risk.

## So we treat it as sensitive

Even de-identified data is protected with behavioural and technological security practices.



**Case · Federal Court of Canada, 2023**

Could partial postal-code (FSA) characters + city names re-identify licensed medical-cannabis growers held by Health Canada?

The court treated these combinable elements as **“personal information”**, exempt from disclosure under the Access to Information Act.

BIAS IN AI / ML

# Models are only as fair as their data

When training data carries bias, models can produce discriminatory outcomes, generalize poorly, and create reputational or legal risk. These biases reflect social and historical inequities embedded through sampling, labeling, and context.



### Sampling

Who is over- or under-represented in the data



### Labeling

Subjective or skewed annotation choices



### Context

The setting in which data was generated

+ **Build responsibly.** Examine how data is collected, labeled and applied. For full requirements see the Data Bias Assessment Guidelines and the associated Data Card.

TYPES OF DATASETS

# Getting data into the DAE



## Your Industry PM is the route

Data is submitted to the DAE through your Industry Project Manager, who coordinates approvals with Vector's Data Office.

*The PM can also guide teams on pre-approved datasets suggested for bootcamp use.*

✓ **PRE-APPROVED LICENSES**

**Apache 2.0 · Creative Commons · MIT**

No Data Submission Form needed if the data also meets public-data criteria and Vector's De-identification Guidelines. Still review attribution, commercial-use, share-alike & derivative terms.

**ALL OTHER LICENSED DATA**

Must be reviewed & approved by the Data Office via a Data Submission Form before transfer. Users are responsible for abiding by every term tied to the data.

# Moving data out: the risk point

## Egress

Transferring data or research products from a controlled environment to an external destination such as cloud services, removable hardware, or an outside party.

**A critical point of vulnerability for leaks, breaches & compliance violations.**

### RULES FOR EGRESS

- ✓ Obtain PM approval for egress (where applicable)
- ✓ Adhere to Vector's Trust & Safety Principles
- ✓ Honour license obligations for all datasets & resources
- ✓ Comply with applicable laws, regulations & agreements

AI TRUST & SAFETY

# Vector's six principles



### Benefit humans & planet

Promote inclusive growth, sustainability & wellbeing



### Reflect democratic values

Uphold human rights, equity & the rule of law



### Respect privacy & security

Handle personal data responsibly; prevent misuse



### Be robust, secure & safe

Monitor & manage across the whole lifecycle



### Responsible disclosure

Outcomes understandable & open to challenge



### Organizational accountability

Stay accountable, backed by oversight

DATA GOVERNANCE

# Policies & procedures

For full detail on Vector's Data Governance Policies and Procedures, see the Vector Wiki.

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PARTICIPANT ACKNOWLEDGEMENT & DISCLAIMER

# Bootcamp terms: the essentials

## KEY DEFINITIONS

### Participant

An individual in the bootcamp, other than an instructor.

### Software

All software, code, applications, materials, datasets & docs provided in, or created within, the bootcamp.

### Vector

Vector Institute, its affiliates, employees, agents and all bootcamp instructors.

## BY PARTICIPATING, YOU AGREE

+ Software is for educational, non-commercial use during the bootcamp only.
+ Any use outside the bootcamp is subject to Vector & third-party license terms, your responsibility to follow.
+ Commercial use or license breach may trigger a claim against you or your organization.
+ Outside use may breach your employer's policies or agreements.
+ Participation transfers no ownership; Vector reserves all rights.

PARTICIPANT ACKNOWLEDGEMENT & DISCLAIMER

# Warranties, liability & governing law



## Disclaimer of warranties

Software is provided "as is" and "as available." Vector disclaims all warranties, express or implied, including merchantability, fitness for purpose, title and non-infringement. No guarantee of error-free or uninterrupted operation.



## Limitation of liability

Vector is not liable for special, incidental, indirect, punitive or consequential damages. Total liability will not exceed the amount paid for the bootcamp, if any, regardless of the cause of action.




## General provisions

This is the entire agreement, superseding prior understandings. Unenforceable terms are reformed minimally; the rest stand. Governed by the laws of Ontario and Canada.

DATA USER OBLIGATIONS



# DAE Compliance Attestation

| | |
|---|---|
| **I WILL** | Access & use granted datasets solely for lawful scientific research, no other purpose. |
| **I WILL NOT** | Use the DAE or its data for personal, commercial, illegal or unethical activity. |
| **I WILL** | Adhere to all Data Transfer Agreement and license requirements for each dataset. |
| **I WILL NOT** | Infringe anyone's intellectual property or related rights, including copyright. |
| **I WILL** | Maintain confidentiality & privacy, preventing unauthorized disclosure. |
| **I WILL** | Implement all required security measures to protect the DAE and datasets. |

# DAE Compliance Attestation

| | |
|---|---|
| **I WILL NOT** | Share login credentials with any other individual. |
| **I WILL NOT** | Interfere with the DAE: no resource monopolizing, malware, unsecured hardware or security tampering. |
| **I WILL NOT** | Export items without supervisor approval; I will check alignment with the Trust & Safety Principles. |
| **I WILL** | Ensure personal-folder datasets meet the public-data criteria, or submit them for review. |
| **I WILL NOT** | Attempt to identify any individual in a de-identified dataset, and will report any such risk found to infosec@vectorinstitute.ai. |
| **I WILL** | Exercise prudent care to avoid disclosing any individual or institution in publications. |

# DAE Compliance Attestation

| | |
|---|---|
| **I UNDERSTAND** | My use may be monitored or audited. Non-compliance can mean loss of access (mine and my supervisor's) and notification to Vector, my institution, funders and regulators. |
| **I WILL** | Include proper attribution to Vector Institute and the data provider for all work using DAE data. |
| **I UNDERSTAND** | It is my responsibility to determine whether REB review is required, and to obtain approval before starting at Vector. |

## Questions?

Reach the Vector Data Office: **dataoffice@vectorinstitute.ai**

RESOURCES

# Training & policy documents

## SUGGESTED TRAINING

- CITI: Good Clinical Practice (GCP)
- CITI: Data/Specimens Only Research (PhysioNet MIMIC)
- Intro to Research Security & Cyber Security for Researchers
- Public Safety Canada: Safeguarding Science Workshop
- General info on Safeguarding Your Research

## POLICIES FOR THIS COURSE

- **Cybersecurity Program**
- **DAE Governance Framework**
- **Data Classification Policy**
- **De-identification Guidelines**
- **Records Retention & Destruction Policy**
- **Technology Asset Access Policy**

RESOURCES

# Key contacts



## Breaches, report immediately

Any privacy or cybersecurity breach, suspected or confirmed, major or minor.

infosec@vectorinstitute.ai



## Technical help with transfers

Support moving data into the Vector DAE.

ops-help@vectorinstitute.ai



## License, agreement & data requests

Questions on dataset terms and any data-related request.

dataoffice@vectorinstitute.ai

# Thank you

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